

Q. 1 (Zzzzzz). Every Monday, Wednesday, and Friday morning you wake up early so you can make it in time to your ORF 309 lecture. As soon as you wake up, you make yourself a cup of coffee, tea, or orange juice with probability 0.5, 0.4, or 0.1, respectively. If you had coffee, tea, or orange juice, then you fall asleep during the lecture with probability 0.1, 0.2, or 0.8, respectively.

- a. Think of your beverage choice/waking status during the lecture on a given class day as the outcome of a random experiment. What is a good sample space for this experiment? Construct the probability measure for this experiment explicitly.
- b. On a given Wednesday, you fall asleep in class. What is the probability that you had orange juice that morning?

Q. 2 (The Chevalier's new game). The Chevalier de Méré is addicted to a new game of chance that has taken the Parisian gambling halls by storm. The game goes as follows: the Chevalier gets to throw a fair, six-sided die; if the number k on the die is odd, the Chevalier wins k dollars; but if the number k is even, the Chevalier loses k dollars. (As usual in financial accounting, we think of losing k dollars as winning $-k$ dollars.)

- a. What is the expected number of dollars the Chevalier wins in one game?
- b. The Chevalier plays 5 games in a row (each game is independent). What is the probability that he loses money in exactly 3 out of the 5 games?

Q. 3 (Guacamole). Avocados are temperamental. When you purchase an avocado, it is hard and inedible. Each day, each avocado ripens independently with probability 0.1 (different avocados and different days are all independent). An avocado can only be eaten on the day that it ripens; by the next day, the avocado is brown and icky.

For the super bowl, you would like to make guacamole. In the morning 2 days before the super bowl, you purchase 5 avocados. (The super bowl is in the evening.)

- a. What is the expected number of days it takes for one avocado to ripen?
- b. What is the expected number of avocados that are ripe during the super bowl? (The avocados start ripening when you purchase them; for the purposes of this question, you can assume they have had 3 full days to ripen by the time the super bowl starts.)
- c. You invite 5 friends to watch the super bowl with you, but the number of friends who actually show up is equally likely to be any number between 0 and 5. Each friend who shows up desires to eat one avocado. What is the probability that exactly the same number of avocados are ripe as the number of friends who show up? (You may assume that your friends and the avocados act independently of each other.)